INCH-POUND
MIL-M-48432B (AR)
25 August 1992
SUPERSEDING
MIL-M-48432A (AR)
31 July 1986

MILITARY SPECIFICATION

MIXER PREAMPLIFIER: 10548160

This specification is approved for use by the U. S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and tests for the Mixer Preamplifier Assembly, 10548160, herein referred to as the mixer preamp, a subassembly of the Microwave Chassis Assembly, 11731190, part of Receiver-Transmitter RT-860/VPS-2, part of Radar Set ANIVPS-2 and Part of Radar Test Set AN/TPM-22 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications standards and handbooks The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in-the solicitation (see 6.2) .

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial COMMENTS (RECOMMENDATIONS, ADDItions,DELETIONS))AND ANY DATA WHICH MAY BE OF USE IN IMPROVING THIS DOCUMENT SHOULD BE ADDRESSED TO: Commander, U.S. Army ARDEC,ATTN: SMCAR-BACS, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DDF0rm1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-109 - **Quality Assurance Terms** and Definitions
MIL-STD-810 - Environmental Test Methods and **Engineering Guidelines**

HANDBOOK

MILITARY

MIL-HDBK-217 - Reliability Stress and Failure Rate Data for Electronic Equipment

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks **are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)**

2.1.2 Other Government documents, drawings and publications **The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.**

DRAWINGS (See 6.4)

U.S.ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10548160 - Mixer Preamplifier
10548182 - Solid State X-Band Source (REF.)
10549175 - Oscillator Module (REF.)
11736113 - Semiconductor Device, Diode, Silicon, Mixer, Matched Pair
12950877 - Requirements for Soldering Electrical **and** Electronic Assemblies

SPECIAL PACKAGING INSTRUCTION

PS **10548160 - packaging of Mixer Preamplifier**

(Copies of drawings, publications and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 10548160, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the mixer preamp shall meet the requirements of MIL-F-13926 as follows:

Dimensions and tolerances.
b. Inorganic protective surface finishings.
Part identification and marking.
d. Workmanship.

3.3 Environmental service conditions.

3*3*1 Vibration. The mixer preamp shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to the equivalent of 6000 miles of track-laying vehicle operational vibration as defined by MIL-STD-810 narrowband random-on-random data for equipment installed on the deck of the Mll3 armored personnel carrier.

3.3.2 Shock. The **mixer preamp shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses** of 15 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude operatihg. The mixer preamp shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg) .

3.3.4 Temperature.

3.3.4.1 Operating. The mixer preamp shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40 **deg F and +155 deg F.**

3.3.4.2 Non-operating (storage). The mixer preamp shall not suffer permanent damage when subjected to the constant temperature extremes of -60 deg F for a Period of 24 hours and +160 deg F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

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3.3.5 Humity (non-operating). The mixer preamp shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 **Electrical** Each conductor shall be electrically continuous as specified on the schematic diagram of the mixer preamp.

3*5 Performance.

3.5.1 **Operation** The mixer preamp shall meet the requirements of this specification while operating into a resistance load impedance Of 50 ± 10 ohms.

3.5.2 **Operating frequencies** The mixer preamp shall operate with a local oscillator frequency of 9145 MHz to 9185 MHz. The signal frequency shall be from 9205 MHz to 9245 MHz. The difference amplifier frequency shall be 60 MHz.

3.5.3 **IF amplifier response** The IF amplifier section shall exhibit -3 dB response points at a bandwidth of 10 ± 2 MHz, centered on the 60 MHz frequency.

3.5.4 Conversion gain. The mixer **preamp shall** measure an overa11 conversion gain of 22 +2 dB -3 dB.

3.5. 5 VSWR. The voltage standing wave ratio, of the mixer preamp, shall not exceed:

Receiver port input 1.20:1 VSWR "

Local **oscillator port input 2.0:1 VSWR**

3.5.6 Loca1 oscillator **power level rquired.** The mixer preamp shall meet all the requirements of this specification with any local oscillator input RF power, to the Lo port, between lmW and 3mW inclusive.

3.5.7 Isolation. The mixer preamp shall exhibit an isolation characteristic of at least -13 dB when comparing the input level of the local oscillator to the power level appearing at the RF flange of the received signal port.

3.5.8 RF input amplitude and gain control.

3.5.8.1 Mixer pream amplification. With the 0.00 ± 0.05 Vdc gain control voltage applied, the mixer preamp shall be capable Of amplifying, with negligible distortion, an RF input signal of -24 dBm (3.9 uW).

3.5.8.2 Mixer pream attenuation. With 6.00 ± 0.25 Vdc gain control voltage applied, the mixer preamp shall be caPable Of attenuating, with negligible distortion, an RF input signal of -5 dBm (.32 mW).

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3.5.8.3 Gain control voltage. AGC gain reduction shall begin at 1.0 ± 0.4 Vdc gain control voltage and shall increase monotonically to 22 +2, -1 dB at 6.00 ± .05 Vdc, following the shape of the curve in Figure 1. The gain control voltage required for any given attenuation shall be within 0.5 Vdc of the curve.

3.5.8.4 Gain control current. With 6.00 ± 0.25 Vdc gain control voltage applied, the gain-control input current shall not exceed 10 ma.

3.5.9 Pulsed gain control. The gain control voltage input waveshape and amplitude shall be as shown in Figure Z. The amplitude shall be 6.0 ± 0.25 Vdc. The gain of the amplifier shall reach a maximum during the period t_1 through t_2 of Figure 2. No spurious outputs in excess of 1.0 mv shall be present.

3.5.10 Noise figure - single sideband. The single sideband noise figure shall be 7.5 dB maximum at room ambient. At +155°F the noise figure shall not exceed 8.5 dB.

3.5.11 Crystals. The crystal diodes used for mixing shall be type 1N23 WGMR per 11736113.

3.5.12 DC power input. The mixer preamp shall be capable of operating from a +28 Vdc ± l percent supply, with a total power consumption of less than 0.7 watts.

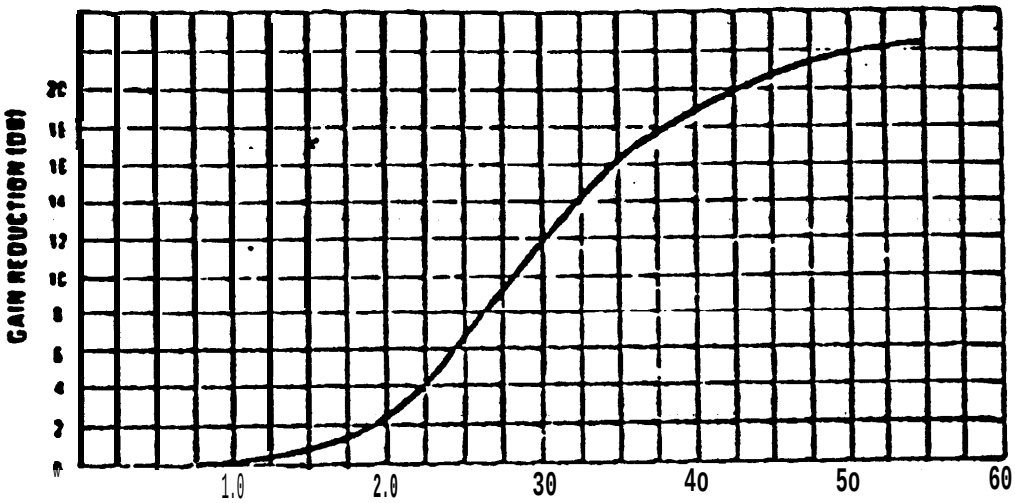
3.5.13 Input filtering. A 25 uF, 50 Vdc rating, capacitor CLR65CJ250MP3 or equal with teflon sleeving over both leads shall be connected inside the mixer preamp case between the +28 Vdc power terminal and case ground.

3.5.14 Pressurization of waveguide ports only. The mixer preamp shall be capable of pressurization to 10 PSIG when bolted to mating connectors and used as a part of a pressurized waveguide system. The leak rate shall be less than 2 PSIG/hour when the sealed volume is 50 cubic inches maximum.

3.6 Reliability. The mean time between failure shall be predicted based upon-procedures and failure rates of MIL-HDBK-217. The vendor shall prepare a complete separate reliability report listing each component, its stress level and the failure rate used for the prediction and submit this report to the procuring activity for approval. Proceeding with the final design shall be contingent upon approval of the contents of this report by the procuring activity reliability department. The bidder shall furnish a preliminary estimate of the MTBF with substantiating calculations per MIL-HDBK-217 as part of his proposal.

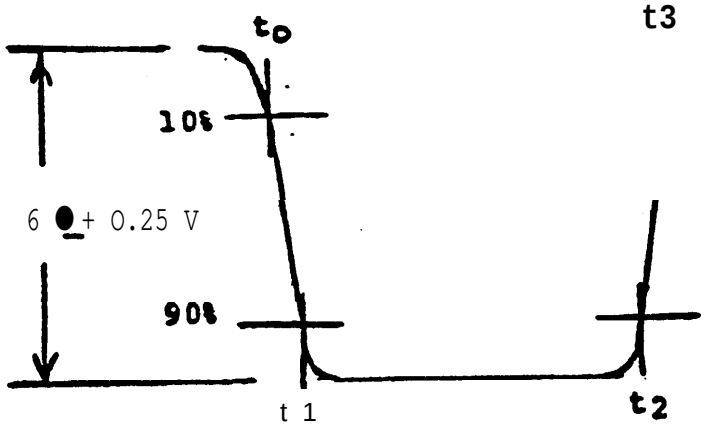
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3.6.1 Minimum mean time between failure. The minimum point estimate MTBF for the Mixer Preamplifier shall be 10,000 hours.



AGC Control Voltage
(Volts DC)

FIGURE 1. Gain vs. control voltage.



t_0 to t_1 = 0.5 µs ±10%

t_0 to t_3 = 7 µs ±10%

t_2 to t_3 = 0.7 µs ±10%

FIGURE 2. Gain control input waveshape.

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3.7 First article When specified in the contract or purchase order, sample shall be-subjected to first article inspection in accordance with the technical provisions herein (see 4.3) .

3.8 Soldering All soldering shall be in accordance with the requirements of Note 2 of Drawing 12950877.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of Sections 3 and 5. The inpection set forth in this specification shall **become** a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of **known defective material, either indicated** or **actual, nor does it commit the Government to accept defective material.**

4.1.2. **General Provisions.** Unless otherwise specified herein, the provisions of MIL-F-13926 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements **specified herein are classified as follows:**

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Special sampling inspection (4.5).

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4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

	Temperature	+60°F to +90°F (+16°C to +32°C)
b.	Relative humidity	90% RH max.
c.	Pressure	28 to 32 inches of Hg

4.3 First article inspection

4.3.1 Sample. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed **below in the quantities indicated.**

Name	Drawing	Quantity
Mixer Preamplifier	10548160	3

4.3.2 **Inspections to be performed. As determined by the Government, the first article-assemblies, components and-test specimens may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.**

4.3.3 Rejection. If any assembly, component, or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of any assembly, component or test specimen to comply with any of the requirements.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation The term 'inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

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4.4.2 Examinations and tests.

4.4.2.1 **Components, assemblies and subassemblies.**

a. The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawings and specifications utilizing as a minimum the conformance criteria specified in this document and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

b. Alternative inspection procedures, methods or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the procuring contracting officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program plan or detailed inspection system, as applicable.

4.4.2.2 Final acceptance inspection. **Subsequent to** first article approval, examinations and tests-related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in Table I shall be performed on a 100% basis.

TABLE 1. Functional tests (loo percent inspection).

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	**None**	**N/A**
Major:		
101. Electrical	**3.4**	**4.7.2**
102. Performance	**3.5**	**4.7.3**
Minor:	None	**N/A**

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4.5 Special sampling One mixer preamp out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table I and shall then meet the tests in Table II. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE II. Special sampling.

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
103. Vibration	3.3.1	4.7.1.1
104. Shock	3.3.2	4.7.1.2
105. Altitude	3.3.3	4.7.1.3
106. Temperature	3.3.4	4.7.1.4
107. Humidity	3.3.5	4.7.1.5
Minor:	None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requireements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination and 'Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment

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is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 **Test equipment** **In conjunction with 4.6.2.1** and 4.6.2.2, the test equipment listed in Table III (Test equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

TABLE 111. **Test equipment list**.

ITEM NUMBER		DESCRIPTION
1.	Power supply dc (2 required)	**O-30 V** at 1 A. **(.05 V** resolution for 0-7 V). Ripple voltage 1.0 **mV maximum.**
2.	Voltmeter, digital	0.1 **mV to 100 Vdc, accuracy 0.1% ± 1 digit.**
3.	Oscilloscope	**Input sensitivity 10 mV/Division. Horizontal deflection 20 ns/Division to 5 s/Division. Dual trace. Vertical amplifier response dc to 150 MHz***
4.	Signal generator	**Frequency range 7 to 11 GHz. Output O dBm to -127 dBm. Output impedance 50 ohms. output cable assembly, 6 feet long, terminated at each end with N male connectors.**
5.	**Microwave generator source**	Frequency range 9140 MHz to 9190 MHz. Output 80 to 120 mW. **Output impedance 50 ohm. Solid state X-band source, part No. (10548182). Crystal osc. 9145 Hz, Dwg No. (10549175)** Crystal osc. 9185 Hz, Dwg No. (10549175)

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TABLE III. Test equipment list. (continued)

ITEM NUMBER	DESCRIPTION
6. Pulse generator	Pulse width 50 ns to 10 ms. output 0.2 volt to 100 volts @50 ohms. Rise time 0.5 us ± 10%. Fall time 0.7 us ± 10%.
7. Power meter and thermistor mount	Power level 10 uW to 10 mW.
8. RF voltmeter, tunable	Frequency range 1 MHz to 1 GHz. Input amplitude range 300 to 1.0 volt. Input impedance 100k ohm shunted with 2.5 pF.
9. Waveguide attenuator	Frequency range 8.2 to 12.4 GHz. Waveguide size (EIA) WR90). Attenuation range 0-50 dB.
10. Coaxial waveguide adapter	Frequency range 8.2 to 12.4 GHz. Waveguide size (EIA) WR90. Coaxial connector, N female.
11. Coaxial adapter TPS plug to BNC jack	Automatic Connector P/N XG551O or equivalent.
12. Dummy load - BNC connector	50 ohm impedance. Frequency range ± 3% dc to 500 MHz. Power rating 2 watts.
13. Directional coupler (2 required)	Frequency range 8.2 to 12.4 GHz. Waveguide size (EIA) WR90. Coupling 10 dB. Directivity 40 dB minimum.
14. Noise generator (matched to item 15)	Waveguide size (EIA) WR90. Frequency range 8 to 12 GHz. Range O to 25 dB.
15. Noise figure indicator	Noise figure range O to 25 dB. Sensitivity 100 microvolt. Frequency range 60 MHz. Input impedance 50 ohms.
16. SWR meter	50 ohm termination. Range 70 dB. Sensitivity 0.15 Vrms to full scale deflection.

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TABLE 111. Test equipment list.. (continued)

Item Number	DESCRIPTION
17. **Frequency counter**	**5 Hz to 80 MHz.** Sensitivity 25 mV rms.
18. Dc millimeter, dual range	Range 0-10 mAdc, 0-100 mAdc, Accuracy 2%.

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test-conditions, general test performance guidance, and individual test methods of MIL-STD-81O, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration **This test shall be performed as specified in Method 514, Category 8, Procedure It of MIL-STD-810 and paragraph 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph 11-2.1 apply. The mixer preamp shall be attached to the vibration generator directly or with a fixture, and securely held by its** normal means of attachment. After vibration, the **mixer preamp shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.**

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-81O except that:

a. **The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.**

b. **The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.**

c. **The maximum g shall occur at 5.5 milliseconds ± 10%.**

d. **The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz.**

After the application of all the shock impulses, the mixer preamp shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude operating **This test shall be performed as specified in Procedure II, Method 500, of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.**

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4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating. This test shall be performed as specified in Procedures I and II; Method 501 of MIL-STD-81O with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating) This test shall be performed as specified in Procedures I and II; Method 502 of MIL-STD-81O with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humity (non-operating) This test shall be performed in accordance with Procedure 111, Method 507 of MIL-STD-81O except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

4.7*3 Performance. The mixer preamp shall be tested using the test set-ups as shown in Figures 3 thru 8 inclusive.

4.7.3.1 Operating Frequencies. Assemble test equipment as shown in Figure 3. verify compliance with 3.5.2.

a. Adjust signal generator dial for a frequency range of approximately 9205 MHz and output level *of* **-10 dBm. Install 9145 MHz crystal oscillator in X-band source. Vary waveguide attenuator until power meter indicates a power level of 1.0 mW at the LO port.**

b. Vary frequency control of signal generator for maximum reading of rf voltmeter. Observe that frequency counter reads 60 MHz.

c. **Duplicate test of 4.6.3.la except replace crystal oscillator** with **assembly marked 9185 MHz and signal generator tuned to 9245 MHz.**

4.7.3.2 If amplifier response. Assemble test equipment as shown in Figure 3. Verify compliance with 3.5.3.

a. Vary the signal generator frequency control while observing the changing level of the rf voltmeter. Note reading of frequency counter when -3 dB points are reached as indicated by the rf voltmeter.

b. Determine bandwidth by calculating difference between the two output frequencies.

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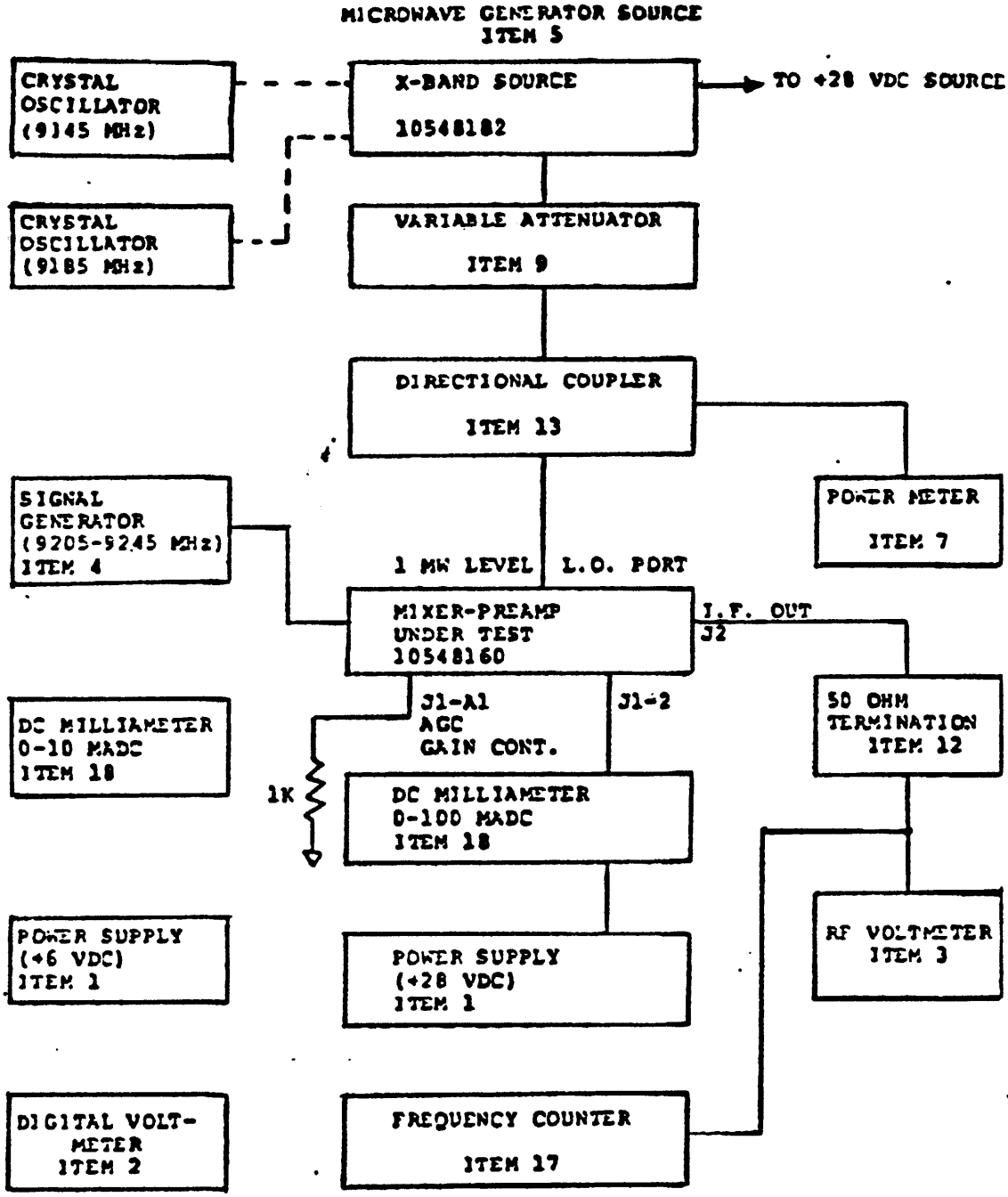


FIGURE 3. Operating frequencies, response and conversion gain test setup.

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4.7.3.3 Conversion gain. Assemble test equipment as shown in Figure 3. Verify compliance with 3.5.4.

a. Vary the signal generator frequency control for maximum reading on the rf voltmeter.

b. Adjust the signal generator output level to -24 dBm. Rf voltmeter shall read between -5 dBm and O dBm.

4.7.3.4 Voltage Standing wave ratio. Assemble test equipment as shown in Figure 4. Verify compliance with 3.5.5.

a. Adjust signal generator for a frequency of 9200 MHz and the output level of O dBm. Connect forward power directional coupler to mixer preamp signal port. Observe VSWR indication.

b. Repeat above test with forward power directional coupler connected to mixer preamp LO port. Observe VSWR indication.

4.7.3.5 Local oscillator power level required. Compliance with 3.5.6 will be evident during performance of tests shown in Figures 3 through 8.

4.7.3.6 Isolation. Assemble test equipment as shown in Figure 5. Verify compliance with 3.5.7.

a. Adjust variable attenuator for a 1 mW level at LO port.

b. Transfer rf power meter to signal port and observe reading. Power meter shall indicate an rf leakage power that is -13 dBm or less.

4.7.3.7 RF dynamic range and gain control. current . ___ Assemble test equipment as shown in Figure 6. verify compliance with 3.5.8.

a. Adjust signal generator to 9205 MHz and output level of -24 dBm. Place switch S1 in position 1. Observe waveform on oscilloscope for absence of distortion.

b. Readjust signal generator output level to -5 dBm and place switch S1 in position 2. Adjust AGC gain control voltage to 6.00 ± 0.25 Vdc. Observe waveform on oscilloscope for absence of distortion.

Observe VOM current drain under conditions of 4.6.3.7b above. Current shall not exceed 10 mA dc. Readjust signal generator output level to -24 dBm. Vary AGC control voltage from 0.0 to 6.0 Vdc in 0.2 volt increments. Gain reduction shall meet the requirements of 3.5.8.3.

d. Observe VOM current drain under conditions of 4.6.3.7b above. Current shall not exceed 10 mA Dc.

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4.7.3.8 Pulsed gain control. **Assemble** test equipment **as** shown in Figure 7. Verify compliance with 3.5.9.

Set controls of pulse generator for negative +6 V to O vdc pulse, pulse width 4.5 us, pulse repetition rate of 20 KHz.

b. Adjust the signal generator for -5 **dBm** input to the mixer preamp. Observe the pulse generator pulse and the **mixer** preamp output level from J2 on dual trace oscilloscope. The gain of the IF amplifier shall reach a maxi-mum during the period t_1 through t_2 of Figure **2.**

4.7.3.9 Figure, single side band. **Assemble** test **equipment as** shown in Figure 8. Verify compliance with 3.5.10. Observe the noise figure on the **VSWR** meter and correct value from double sideband to single sideband by subtracting 3.3 dB from this reading.

4.7.3.10 DC **Observe reading of VOM (0-100 mA) dc meter with test equipment assembled as shown in Figure 3. DC current shall not exceed 25 mA.**

4.7.3.11 Input filtering Verify visually that the filtering capacitor has been installed/connected properly as specified in 3.5.13.

4.7.3.12 Pressurization of wave guide ports only **Observe reading on pressure gauge (O - 10 PSIG) over a one hour period, the total pressure** drop shall be less than 2 PSIG as per 3.5.14.

4.8 Reliability **A reliability demonstration test shall be conducted as part of FAT/IPT. A minimum of 20 hours of operation on each sample (sample size 6** minimum) shall be accumulated during/ after exposure to each environment of this specification (see 3.3) . Additional hours are required at 75° ± 15°F in order to accumulate a minimum test time of 10,000 hours. Testing shall continue until a minimum MTBF (point estimate) of 10,000 hours is demonstrated or until 40,000 hours of test time have been accomplished.

4.8.1 Data collection. Data collected will consist of all equipment malfunctions, number of hours completed when malfunction occurred, nature and cause of malfunction, time required to correct malfunction, and required corrective action.

4.9 **Soldering** Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of Note 2 of Drawing 12950877 as specified in paragraph 3.8.

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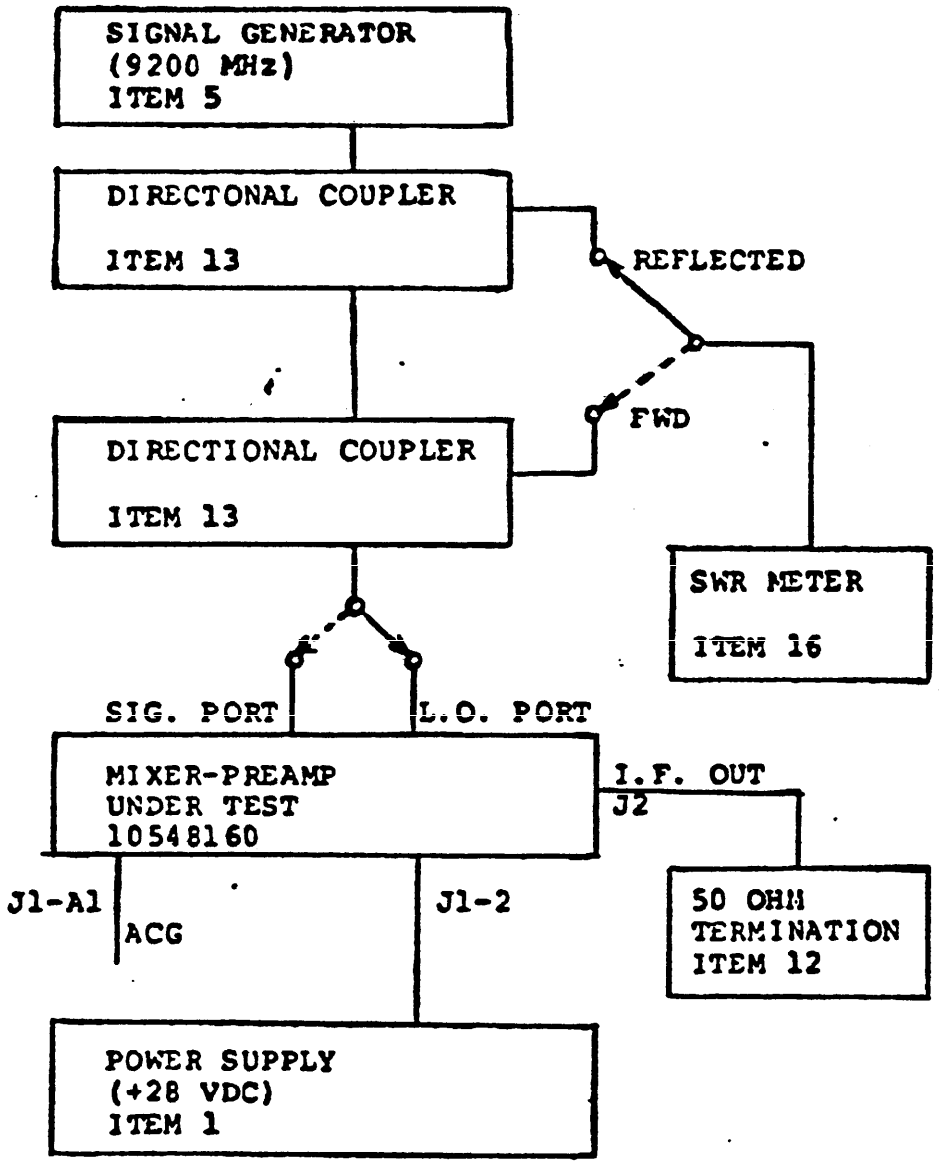


FIGURE 4. VSWR setup.

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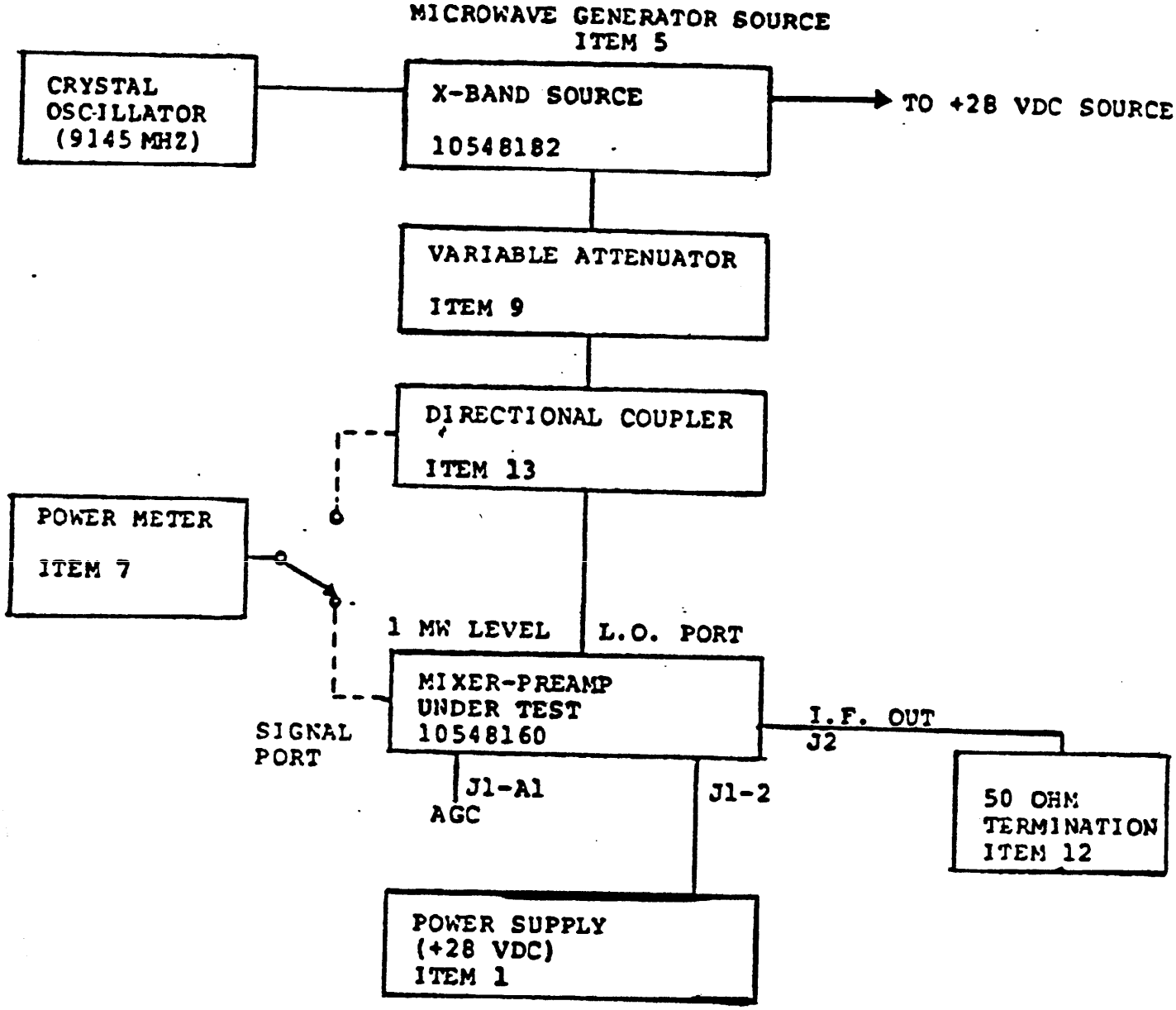


FIGURE 5. Isolation test setup.

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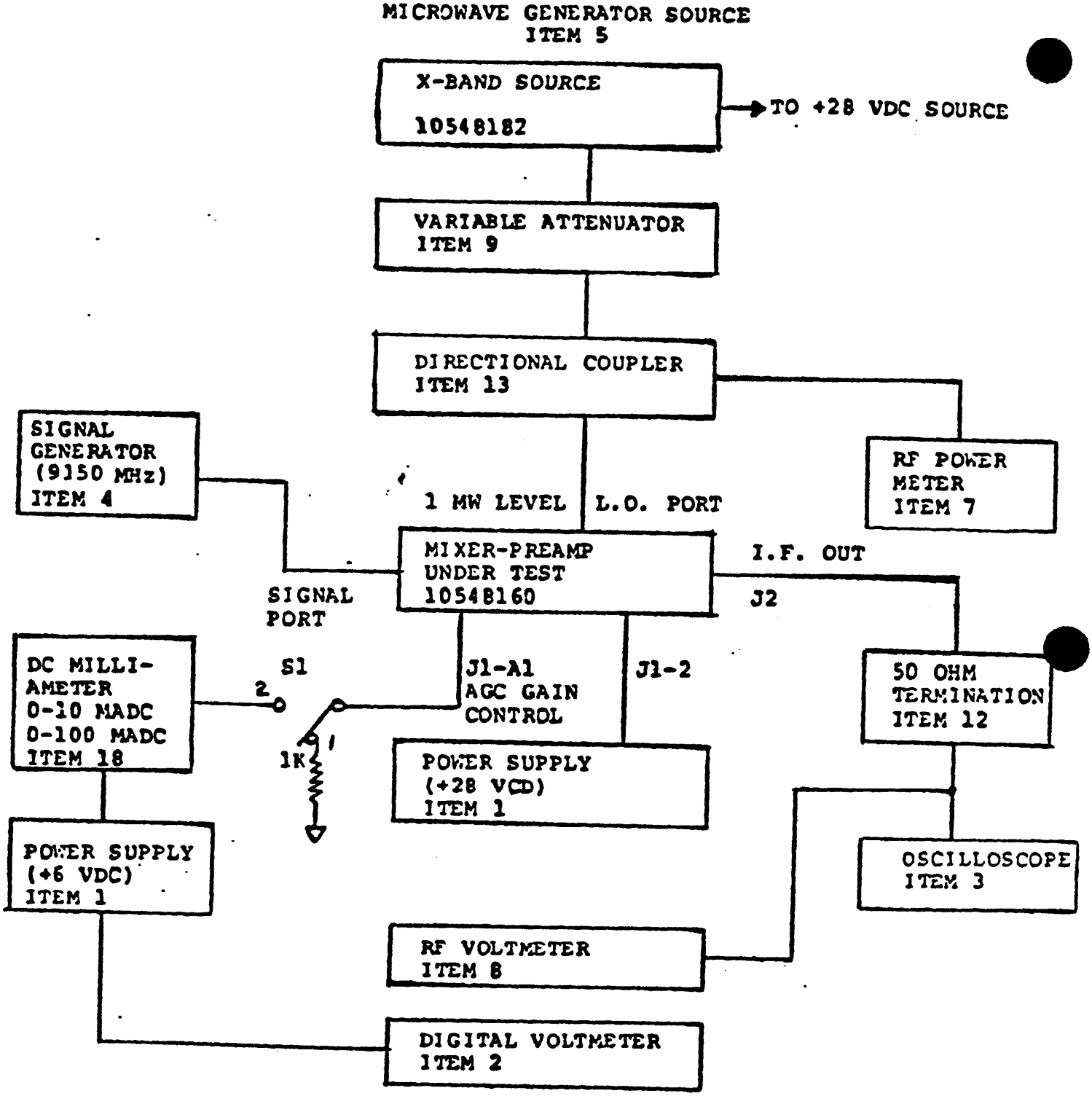


FIGURE 6. RF dynamic range and gain control test setup.

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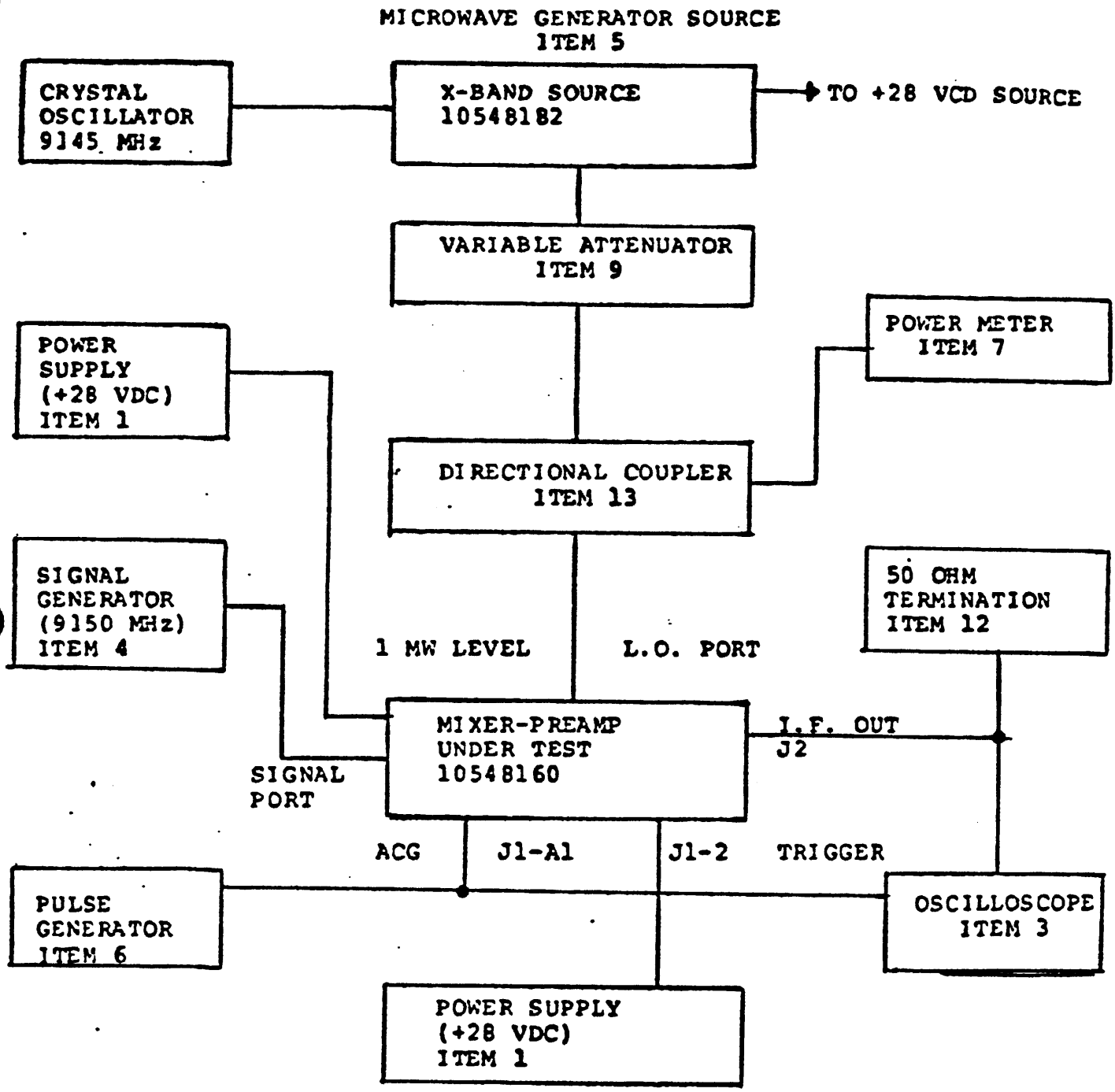


FIGURE 7. Pulsed gain control test setup.

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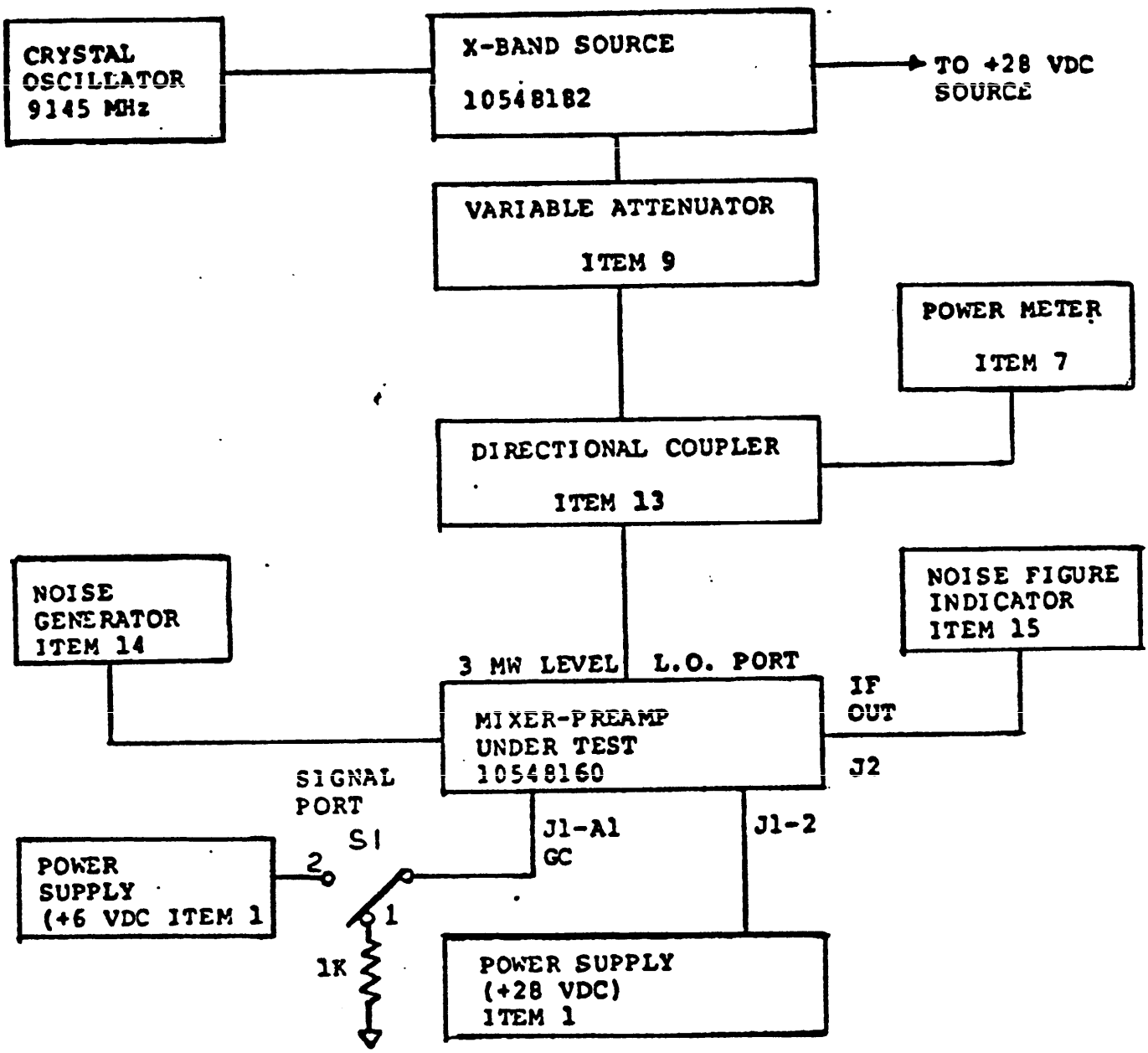


FIGURE 8. Noise figure test setup.

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5. PACKAGING

5.1 Preservation, packing and marking shall be in accordance with Special Packaging Instruction PS 10548160. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The mixer preamp is required to combine the received radar signal and the output of the receiver local oscillator. This signal is amplified and produces the 60 MHz i-f voltage supplied to Receiver-Transmitter RT-860/VPS-2.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

c. Applicable stock number.

d. First article sample requirements.

e. Packaging requirements, if other than specified in Section 5.

f. Serialization requirements, if applicable.

g. Certification of conformance for each lot of shipment of product.

6.3 Submission of contractor inspection equipment designs for Approval. Submit two copies of designs as required to: Commander, U.S. Army ARDEC, Attn: SMCAR-QAF-1, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Drawings Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

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6.5 Subject term (key word) listing..

Microwave Chassis
Radar Set AN/VPS-2
Receiver-Transmitter RT-860/VPS-2

6.6 Changes from previous issue **Asterisks (or vertical lines) are not used in this revision to identify changes with respect to** the previous **issue** due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1285-A740)

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	MIL-M-48432	25 AUGUST 1992

3. DOCUMENT TITLE: MIXER PREAMPLIFIER: 10548160

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	e. DATE SUBMITTED (YYMMDD)

7. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)
U.S ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial: (201) 724-6674 (2) AUTOVON: 880-6674
c. ADDRESS (Include Zip Code) ATTN: SMACA-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 — *Previous editions are obsolete.* — 198/290

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